Supplementary Materials

Synergies of electrical and sectoral integration: Analysing geographical multi-node scenarios with sector coupling variations for a transition towards a fully renewables-based energy system

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Section S1. Financial and technical assumptions.

Table S1.1 Financial and technical assumptions of energy system technologies used.

Technology		Unit	2020	2025	2030	2035	2040	2045	2050	Sources
PV fixed tilted PP	Capex	€/kW$_{el}$	432	336	278	237	207	184	166	[1–3]
	Opex fix	€/(kW$_{el}$ a)	7.76	6.51	5.66	5	4.47	4.04	3.7	
	Opex var	€/kWel	0	0	0	0	0	0	0	
	Lifetime	years	30	35	35	35	40	40	40	
PV rooftop - residential	Capex	€/kW$_{el}$	1045	842	715	622	551	496	453	[1,3]
	Opex fix	€/(kW$_{el}$ a)	9.13	7.66	6.66	5.88	5.26	4.75	4.36	
	Opex var	€/kWh$_{el}$	0	0	0	0	0	0	0	
	Lifetime	years	30	35	35	35	40	40	40	
PV rooftop - commercial	Capex	€/kW$_{el}$	689	544	456	393	345	308	280	[1,3]
	Opex fix	€/(kW$_{el}$ a)	9.13	7.66	6.66	5.88	5.26	4.75	4.36	
	Opex var	€/kWh$_{el}$	0	0	0	0	0	0	0	
	Lifetime	years	30	35	35	35	40	40	40	

PV rooftop - industrial	Capex	€/kW_{el}	512	397	329	281	245	217	197	[1,3]
	Opex fix	€/(kW_{el} a)	9.13	7.66	6.66	5.88	5.26	4.75	4.36	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	30	35	35	35	40	40	40	
PV single-axis PP	Capex	€/kW_{el}	475	370	306	261	228	202	183	[2–4]
	Opex fix	€/(kW_{el} a)	8.54	7.16	6.23	5.5	4.92	4.44	4.07	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	30	35	35	35	40	40	40	
Wind onshore PP	Capex	€/kW_{el}	1150	1060	1000	965	940	915	900	[5–7]
	Opex fix	€/(kW_{el} a)	23	21.2	20	19.3	18.8	18.3	18	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	25	25	25	25	
Hydro Run-of-River PP	Capex	€/kW_{el}	2560	2560	2560	2560	2560	2560	2560	[8]
	Opex fix	€/(kW_{el} a)	76.8	76.8	76.8	76.8	76.8	76.8	76.8	
	Opex var	€/kWh_{el}	0.005	0.005	0.005	0.005	0.005	0.005	0.005	
	Lifetime	years	50	50	50	50	50	50	50	
Hydro Reservoir/ Dam	Capex	€/kW_{el}	1650	1650	1650	1650	1650	1650	1650	[8]
	Opex fix	€/(kW_{el} a)	49.5	49.5	49.5	49.5	49.5	49.5	49.5	
	Opex var	€/kWh_{el}	0.003	0.003	0.003	0.003	0.003	0.003	0.003	
	Lifetime	years	50	50	50	50	50	50	50	
Geothermal PP	Capex	€/kW_{el}	4970	4720	4470	4245	4020	3815	3610	[8,9]
	Opex fix	€/(kW_{el} a)	80	80	80	80	80	80	80	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	40	40	40	40	40	40	40	
Steam turbine (CSP)	Capex	€/kW_{el}	968	946	923	902	880	860	840	[10,11]
	Opex fix	€/(kW_{el} a)	19.4	18.9	18.5	18	17.6	17.2	16.8	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	30	30	30	30	

CCGT PP	Capex	€/kW$_{el}$	775	775	775	775	775	775	775	[12]
	Opex fix	€/(kW$_{el}$ a)	19.375	19.375	19.375	19.375	19.375	19.375	19.375	
	Opex var	€/kWh$_{el}$	0.002	0.002	0.002	0.002	0.002	0.002	0.002	
	Lifetime	years	35	35	35	35	35	35	35	
OCGT PP	Capex	€/kW$_{el}$	475	475	475	475	475	475	475	[7,13]
	Opex fix	€/(kW$_{el}$ a)	14.25	14.25	14.25	14.25	14.25	14.25	14.25	
	Opex var	€/kWh$_{el}$	0.011	0.011	0.011	0.011	0.011	0.011	0.011	
	Lifetime	years	35	35	35	35	35	35	35	
Internal Combustion Generator	Capex	€/kW$_{el}$	385	385	385	385	385	385	385	[6,14]
	Opex fix	€/(kW$_{el}$ a)	11.5	11.5	11.5	11.5	11.5	11.5	11.5	
	Opex var	€/kWh$_{el}$	0.0047	0.0047	0.0047	0.0047	0.0047	0.0047	0.0047	
	Lifetime	years	30	30	30	30	30	30	30	
Coal PP	Capex	€/kW$_{el}$	1500	1500	1500	1500	1500	1500	1500	[7,15,16]
	Opex fix	€/(kW$_{el}$ a)	20	20	20	20	20	20	20	
	Opex var	€/kWh$_{el}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	
	Lifetime	years	40	40	40	40	40	40	40	
Biomass PP	Capex	€/kW$_{el}$	2620	2475	2330	2195	2060	1945	1830	[7,17,18]
	Opex fix	€/(kW$_{el}$ a)	47.2	44.6	41.9	39.5	37.1	35	32.9	
	Opex var	€/kWh$_{el}$	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	
	Lifetime	years	25	25	25	25	25	25	25	
Nuclear PP	Capex	€/kW$_{el}$	6003	6003	5658	5658	5244	5244	5175	[7,15,19]
	Opex fix	€/(kW$_{el}$ a)	113.1	113.1	98.4	98.4	83.6	83.6	78.8	
	Opex var	€/kWh$_{el}$	0.0025	0.0025	0.0025	0.0025	0.0025	0.0025	0.0025	
	Lifetime	years	40	40	40	40	40	40	40	
CHP NG Heating	Capex	€/kW$_{el}$	880	880	880	880	880	880	880	[7,17,18]
	Opex fix	€/(kW$_{el}$ a)	74.8	74.8	74.8	74.8	74.8	74.8	74.8	
	Opex var	€/kWh$_{el}$	0.0024	0.0024	0.0024	0.0024	0.0024	0.0024	0.0024	
	Lifetime	years	30	30	30	30	30	30	30	
	Capex	€/kW$_{el}$	880	880	880	880	880	880	880	[7,17,18]

CHP Oil Heating	Opex fix	€/(kW$_{el}$ a)	74.8	74.8	74.8	74.8	74.8	74.8	74.8	
	Opex var	€/kWh$_{el}$	0.0024	0.0024	0.0024	0.0024	0.0024	0.0024	0.0024	
	Lifetime	years	30	30	30	30	30	30	30	
CHP Coal Heating	Capex	€/kW$_{el}$	2030	2030	2030	2030	2030	2030	2030	[7,17,18]
	Opex fix	€/(kW$_{el}$ a)	46.7	46.7	46.7	46.7	46.7	46.7	46.7	
	Opex var	€/kWh$_{el}$	0.0051	0.0051	0.0051	0.0051	0.0051	0.0051	0.0051	
	Lifetime	years	40	40	40	40	40	40	40	
CHP Biomass Heating	Capex	€/kW$_{el}$	3400	3300	3200	3125	3050	2975	2900	[17,18,20]
	Opex fix	€/(kW$_{el}$ a)	97.6	94.95	92.3	90.8	89.3	87.8	86.3	
	Opex var	€/kWh$_{el}$	0.0038	0.0038	0.0037	0.0037	0.0038	0.0038	0.0038	
	Lifetime	years	25	25	25	25	25	25	25	
CHP Biogas	Capex	€/kW$_{el}$	429.2	399.6	370	340.4	325.6	310.8	296	[17,18]
	Opex fix	€/(kW$_{el}$ a)	17.168	15.984	14.8	13.616	13.024	12.432	11.84	
	Opex var	€/kWh$_{el}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	
	Lifetime	years	30	30	30	30	30	30	30	
MSW incinerator	Capex	€/kW$_{el}$	5630	5440	5240	5030	4870	4690	4540	[7,17,18]
	Opex fix	€/(kW$_{el}$ a)	253.35	244.8	235.8	226.35	219.15	211.05	204.3	
	Opex var	€/kWh$_{el}$	0.0069	0.0069	0.0069	0.0069	0.0069	0.0069	0.0069	
	Lifetime	years	30	30	30	30	30	30	30	
Concentrating solar thermal (CSP) – solar field, parabolic trough	Capex	€/kW$_{th}$	344.5	303.6	274.7	251.1	230.2	211.9	196	[10,11,21,22]
	Opex fix	€/(kW$_{th}$ a)	7.9	7	6.3	5.8	5.3	4.9	4.5	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	25	25	25	25	
Residential Solar Heat - Collectors - Space Heating	Capex	€/kW$_{th}$	1214	1179	1143	1071	1000	929	857	[17,18]
	Opex fix	€/(kW$_{th}$ a)	14.8	14.8	14.8	14.8	14.8	14.8	14.8	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	25	25	30	30	30	30	30	
Residential Solar Heat -	Capex	€/kW$_{th}$	485	485	485	485	485	485	485	[17,18]
	Opex fix	€/(kW$_{th}$ a)	4.85	4.85	4.85	4.85	4.85	4.85	4.85	

Collectors - Hot Water	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	15	15	15	15	15	15	15	
DH Electric Heating	Capex	€/kW$_{th}$	100	100	75	75	75	75	75	[17,18]
	Opex fix	€/(kW$_{th}$ a)	1.47	1.47	1.47	1.47	1.47	1.47	1.47	
	Opex var	€/kWh$_{th}$	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	
	Lifetime	years	35	35	35	35	35	35	35	
DH Heat Pump	Capex	€/kW$_{th}$	660	618	590	568	554	540	530	[17,18,20]
	Opex fix	€/(kW$_{th}$ a)	2	2	2	2	2	2	2	
	Opex var	€/kWh$_{th}$	0.0018	0.0017	0.0017	0.0016	0.0016	0.0016	0.0016	
	Lifetime	years	25	25	25	25	25	25	25	
DH NG Heating	Capex	€/kW$_{th}$	75	75	100	100	100	100	100	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2.775	2.775	3.7	3.7	3.7	3.7	3.7	
	Opex var	€/kWh$_{th}$	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	
	Lifetime	years	35	35	35	35	35	35	35	
DH Oil Heating	Capex	€/kW$_{th}$	75	75	100	100	100	100	100	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2.775	2.775	3.7	3.7	3.7	3.7	3.7	
	Opex var	€/kWh$_{th}$	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	
	Lifetime	years	35	35	35	35	35	35	35	
DH Coal Heating	Capex	€/kW$_{th}$	75	75	100	100	100	100	100	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2.775	2.775	3.7	3.7	3.7	3.7	3.7	
	Opex var	€/kWh$_{th}$	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	
	Lifetime	years	35	35	35	35	35	35	35	
DH Biomass Heating	Capex	€/kW$_{th}$	75	75	100	100	100	100	100	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2.8	2.8	3.7	3.7	3.7	3.7	3.7	
	Opex var	€/kWh$_{th}$	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	
	Lifetime	years	35	35	35	35	35	35	35	
DH Geothermal Heat	Capex	€/kW$_{th}$	3642	3384	3200	3180	3160	3150	3146	[17,18,23]
	Opex fix	€/(kW$_{th}$ a)	133	124	117	116	115	115	115	

	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	22	22	22	22	22	22	22	
Local Electric Heating	Capex	€/kW$_{th}$	100	100	100	100	100	100	100	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2	2	2	2	2	2	2	
	Opex var	€/kWh$_{th}$	0.001	0.001	0.001	0.001	0.001	0.001	0.001	
	Lifetime	years	30	30	30	30	30	30	30	
Local Heat Pump	Capex	€/kW$_{th}$	780	750	730	706	690	666	650	[7,17,18]
	Opex fix	€/(kW$_{th}$ a)	15.6	15	7.3	7.1	6.9	6.7	6.5	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Local NG Heating	Capex	€/kW$_{th}$	800	800	800	800	800	800	800	[17,18]
	Opex fix	€/(kW$_{th}$ a)	27	27	27	27	27	27	27	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	22	22	22	22	22	22	22	
Local Oil Heating	Capex	€/kW$_{th}$	440	440	440	440	440	440	440	[17,18]
	Opex fix	€/(kW$_{th}$ a)	18	18	18	18	18	18	18	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Local Biomass Heating	Capex	€/kW$_{th}$	675	675	750	750	750	750	750	[17,18]
	Opex fix	€/(kW$_{th}$ a)	2	2	3	3	3	3	3	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Local Biogas Heating	Capex	€/kW$_{th}$	800	800	800	800	800	800	800	[17,18]
	Opex fix	€/(kW$_{th}$ a)	27	27	27	27	27	27	27	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	22	22	22	22	22	22	22	
Water Electrolysis	Capex	€/kW$_{H2}$	685	500	380	325	296	267	248	[24,25]
	Opex fix	€/(kW$_{H2}$ a)	23.98	17.5	13.3	11.38	10.36	9.35	8.68	
	Opex var	€/kWh$_{H2}$	0.0012	0.0012	0.0012	0.0012	0.0012	0.0012	0.0012	

	Lifetime	years	30	30	30	30	30	30	30	
CO_2 direct air capture	Capex	€/(t_{CO2} a)	730	481	338	281	237	217	199	[26,27]
	Opex fix	€/(t_{CO2} a)	29.2	19.2	13.5	11.2	9.5	8.7	8	
	Opex var	€/t_{CO2}	0	0	0	0	0	0	0	
	Lifetime	years	20	30	25	30	30	30	30	
Methanation	Capex	€/kW_{SNG}	502	368	278	247	226	204	190	[21,24,25]
	Opex fix	€/(kW_{SNG} a)	23.09	16.93	12.79	11.36	10.4	9.38	8.74	
	Opex var	€/MWh_{SNG}	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	
	Lifetime	years	30	30	30	30	30	30	30	
Biogas digester	Capex	€/kW_{th}	730.61	705.95	680	652.75	631.98	608.63	589.16	[17,18]
	Opex fix	€/(kW_{th} a)	29.22	28.24	27.2	26.11	25.28	24.35	23.57	
	Opex var	€/kWh_{th}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	25	25	25	25	
Biogas Upgrade	Capex	€/kW_{th}	290	270	250	230	220	210	200	[17,18]
	Opex fix	€/(kW_{th} a)	23.2	21.6	20	18.4	17.6	16.8	16	
	Opex var	€/kWh_{th}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Fischer-Tropsch unit	Capex	€/kW,FT_{Liq}	947	947	947	947	852.3	852.3	852.3	[17,18]
	Opex fix	€/kW,FT_{Liq}	28.41	28.41	28.41	28.41	25.57	25.57	25.57	
	Opex var	€/kWh,FT_{Liq}	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	
Gas Liquefaction	Capex	€/kW_{Liq}	181.1	181.1	181.1	181.1	181.1	181.1	181.1	[17,18]
	Opex fix	€/kW_{Liq}	6.34	6.34	6.34	6.34	6.34	6.34	6.34	
	Opex var	€/kWh_{Liq}	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	25	25	25	25	
H_2 Liquefaction	Capex	€/kW_{Liq}	358.1	358.1	358.1	175.9	152.9	145.2	137.9	[28–30]
	Opex fix	€/kW_{Liq}	14.32	14.32	14.32	7.03	6.11	5.81	5.52	
	Opex var	€/kWh_{Liq}	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	

Steam Methane Reforming	Capex	€/kW_{H2}	320	320	320	320	320	320	320	[31]
	Opex fix	€/kW_{H2}	16	16	16	16	16	16	16	
	Opex var	€/kWh_{H2}	0	0	0	0	0	0	0	
	Lifetime	years	30	30	30	30	30	30	30	
Battery Storage	Capex	€/kWh_{el}	234	153	110	89	76	68	61	[2,11,32,33]
	Opex fix	€/(kWh_{el} a)	3.28	2.6	2.2	2.05	1.9	1.77	1.71	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery Interface	Capex	€/kW_{el}	117	76	55	44	37	33	30	[11,32–36]
	Opex fix	€/(kW_{el} a)	1.64	1.29	1.1	1.01	0.93	0.86	0.84	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery PV prosumer - residential Storage	Capex	€/kWh_{el}	462	308	224	182	156	140	127	[2,17,18,33]
	Opex fix	€/(kWh_{el} a)	5.08	4	3.36	3.09	2.81	2.8	2.54	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery PV prosumer - residential Interface	Capex	€/kW_{el}	231	153	112	90	76	68	62	[2,17,18,33]
	Opex fix	€/(kW_{el} a)	2.54	1.99	1.68	1.53	1.37	1.36	1.24	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery PV prosumer - commercial Storage	Capex	€/kWh_{el}	366	240	175	141	121	108	98	[2,17,18,33]
	Opex fix	€/(kWh_{el} a)	4.39	3.6	2.98	2.68	2.54	2.38	2.25	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery PV prosumer - commercial Interface	Capex	€/kW_{el}	183	119	88	70	59	53	48	[2,17,18,33]
	Opex fix	€/(kW_{el} a)	2.2	1.79	1.5	1.33	1.24	1.17	1.1	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
	Capex	€/kWh_{el}	278	181	131	105	90	80	72	[2,17,18,33]

Battery PV prosumer - industrial Storage	Opex fix	€/(kWh_{el} a)	3.89	3.08	2.62	2.42	2.25	2.08	1.94	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
Battery PV prosumer - industrial Interface	Capex	€/kW_{el}	139	90	66	52	44	39	35	[2,17,18,33]
	Opex fix	€/(kW_{el} a)	1.95	1.53	1.32	1.2	1.1	1.01	0.95	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	20	20	20	20	20	20	20	
PHES Storage	Capex	€/kWh_{el}	7.7	7.7	7.7	7.7	7.7	7.7	7.7	[7,8,37]
	Opex fix	€/(kWh_{el} a)	1.335	1.335	1.335	1.335	1.335	1.335	1.335	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	50	50	50	50	50	50	50	
PHES Interface	Capex	€/kW_{el}	650	650	650	650	650	650	650	[7,8,37]
	Opex fix	€/(kW_{el} a)	0	0	0	0	0	0	0	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	50	50	50	50	50	50	50	
A-CAES Storage	Capex	€/kWh_{el}	75	65.3	57.9	53.6	50.8	47	43.8	[7,8,18]
	Opex fix	€/(kWh_{el} a)	1.16	0.99	0.87	0.81	0.77	0.71	0.66	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	40	40	40	40	40	40	40	
A-CAES Interface	Capex	€/kW_{el}	540	540	540	540	540	540	540	[7,8,18]
	Opex fix	€/(kW_{el} a)	17.5	17.5	17.5	17.5	17.5	17.5	17.5	
	Opex var	€/kWh_{el}	0	0	0	0	0	0	0	
	Lifetime	years	40	40	40	40	40	40	40	
Thermal energy storage	Capex	€/kWh_{th}	41.8	32.7	26.8	23.3	21	19.3	17.5	[17,18]
	Opex fix	€/(kWh_{th} a)	0.63	0.49	0.4	0.35	0.32	0.29	0.26	
	Opex var	€/kWh_{th}	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	30	30	30	30	
Thermal energy storage	Capex	€/kW_{th}	0	0	0	0	0	0	0	[17,18]
	Opex fix	€/(kW_{th} a)	0	0	0	0	0	0	0	

Interface	Opex var	€/kW$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	30	30	30	30	
Hydrogen Storage	Capex	€/kWh$_{th}$	0.24	0.24	0.24	0.24	0.24	0.24	0.24	[38,39]
	Opex fix	€/(kWh$_{th}$ a)	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	
	Opex var	€/kWh$_{th}$	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	
	Lifetime	years	30	30	30	30	30	30	30	
Hydrogen Storage Interface	Capex	€/kW$_{th}$	100	100	100	100	100	100	100	[38,39]
	Opex fix	€/(kW$_{th}$ a)	4	4	4	4	4	4	4	
	Opex var	€/kW$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	15	15	15	15	15	15	15	
CO_2 Storage	Capex	€/ton	142	142	142	142	142	142	142	[26]
	Opex fix	€/(ton a)	9.94	9.94	9.94	9.94	9.94	9.94	9.94	
	Opex var	€/ton	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	
	Lifetime	years	30	30	30	30	30	30	30	
CO_2 Storage Interface	Capex	€/ton/h	0	0	0	0	0	0	0	[26]
	Opex fix	€/(ton/h a)	0	0	0	0	0	0	0	
	Opex var	€/ton	0	0	0	0	0	0	0	
	Lifetime	years	50	50	50	50	50	50	50	
Gas Storage	Capex	€/kWh$_{th}$	0.05	0.05	0.05	0.05	0.05	0.05	0.05	[39]
	Opex fix	€/(kWh$_{th}$ a)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	50	50	50	50	50	50	50	
Gas Storage Interface	Capex	€/kW$_{th}$	100	100	100	100	100	100	100	[39]
	Opex fix	€/(kW$_{th}$ a)	4	4	4	4	4	4	4	
	Opex var	€/kW$_{th}$	0	0	0	0	0	0	0	
	Lifetime	years	15	15	15	15	15	15	15	
District Heat Storage	Capex	€/kWh$_{th}$	40	30	30	25	20	20	20	[17,18]
	Opex fix	€/(kWh$_{th}$ a)	0.6	0.45	0.45	0.375	0.3	0.3	0.3	
	Opex var	€/kWh$_{th}$	0	0	0	0	0	0	0	

	Lifetime	years	25	25	25	30	30	30	30	
District Heat Storage Interface	Capex	€/kW_{th}	0	0	0	0	0	0	0	[17,18]
	Opex fix	€/(kW_{th} a)	0	0	0	0	0	0	0	
	Opex var	€/kW_{th}	0	0	0	0	0	0	0	
	Lifetime	years	25	25	25	30	30	30	30	
Reverse Osmosis Seawater Desalination	Capex	€/(m^3/day)	960	835	725	630	550	480	415	[40]
	Opex fix	€/(m^3/day)	38.4	33.4	29	25.2	22	19.2	16.6	
	Consumption	kWh_{th}/m^3	0	0	0	0	0	0	0	
	Lifetime	years	25	30	30	30	30	30	30	
	Consumption	kWh_{el}/m^3	3.6	3.35	3.15	3	2.85	2.7	2.6	
Multi-Stage Flash Stand-alone	Capex	€/(m^3/day)	2000	2000	2000	2000	2000	2000	2000	[40]
	Opex fix	€/(m^3/day)	100	100	100	100	100	100	100	
	Consumption	kWh_{th}/m^3	85	85	85	85	85	85	85	
	Lifetime	years	25	25	25	25	25	25	25	
	Consumption	kWh_{el}/m^3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	
Multi Stage Flash Cogeneration	Capex	€/(m^3/day)	3069	3069	3069	3069	3069	3069	3069	[17,18]
	Opex fix	€/(m^3/day)	121.4	121.4	121.4	121.4	121.4	121.4	132.1	
	Consumption	kWh_{th}/m^3	202.5	202.5	202.5	202.5	202.5	202.5	202.5	
	Lifetime	years	25	25	25	25	25	25	25	
	Consumption	kWh_{el}/m^3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	
Multi Effect Distillation Stand-alone	Capex	€/(m^3/day)	1200	1044	906.3	787.5	687.5	600	518.8	[40]
	Opex fix	€/(m^3/day)	39.6	34.44	29.91	25.99	22.69	19.8	17.12	
	Consumption	kWh_{th}/m^3	51	44	38	32	28	28	28	
	Lifetime	years	25	25	25	25	25	25	25	
	Consumption	kWh_{el}/m^3	1.5	1.5	1.5	1.5	1.5	1.5	1.5	
Multi Effect Distillation Cogeneration	Capex	€/(m^3/day)	2150	2150	2150	2150	2150	2150	2150	[17,18]
	Opex fix	€/(m^3/day)	61.69	61.69	61.69	61.69	61.69	61.69	68.81	
	Consumption	kWh_{th}/m^3	168	168	168	168	168	168	168	
	Lifetime	years	25	25	25	25	25	25	25	

	Consumption	kWh_{el}/m^3	1.5	1.5	1.5	1.5	1.5	1.5	1.5	
Water Storage	Capex	€/m^3	64.59	64.59	64.59	64.59	64.59	64.59	64.59	[40]
	Opex fix	€/m^3	1.29	1.29	1.29	1.29	1.29	1.29	1.29	
	Opex var	€/m^3	0	0	0	0	0	0	0	
	Lifetime	years	50	50	50	50	50	50	50	

Table S1.2 Financial assumptions for fossil-nuclear fuels and CO_2 emissions.

Component	Unit	2020	2025	2030	2035	2040	2045	2050	Sources
Coal	€/MWh_{th}	7.7	8.4	9.2	10.2	11.1	11.1	11.1	[38]
Fuel Oil	€/MWh_{th}	84.8	94.0	103.2	102.2	101.3	101.3	101.3	[15]
Fossil gas	€/MWh_{th}	22.2	30.0	32.7	36.1	40.2	40.2	40.2	[38]
Uranium	€/MWh_{th}	2.6	2.6	2.6	2.6	2.6	2.6	2.6	[41]
CO_2 emissions	€/tCO_2	28	52	61	68	75	100	150	[38]
GHG emissions by fuel type									
Coal		tCO_{2eq}/MWh_{th}			0.34		[42]		
Oil		tCO_{2eq}/MWh_{th}			0.28		[42]		
Fossil gas		tCO_{2eq}/MWh_{th}			0.24		[42]		

Table S1.3 Efficiency assumptions for HVDC and HVAC transmission.

Component	Power losses	Source
HVDC line	1.6% / 1000 km	[43]
HVDC converter pair	1.4%	[43]
HVAC line	9.4% / 1000 km	[43]

Table S1.4 Electricity retail prices for residential consumers by zone.

Zone	Unit	2015[1]	2020	2025	2030	2035	2040	2045	2050
CL-N	[€/kWh]	0.161	0.187	0.216	0.249	0.270	0.283	0.298	0.313
CL-CN	[€/kWh]	0.182	0.210	0.242	0.261	0.281	0.295	0.310	0.326
CL-C	[€/kWh]	0.158	0.184	0.213	0.243	0.267	0.281	0.295	0.310
CL-S	[€/kWh]	0.185	0.215	0.247	0.267	0.280	0.295	0.310	0.326
CL-A	[€/kWh]	0.240	0.268	0.281	0.296	0.311	0.327	0.343	0.361
CL-M	[€/kWh]	0.178	0.206	0.239	0.266	0.280	0.294	0.309	0.325

[1] These values are the average annual prices of the year 2018, obtained from the website of each retail company that operates as an electricity distributor in the Chilean regional capitals.

Table S1.5 Electricity retail prices for commercial consumers by zone.

Zone	Unit	2015[1]	2020	2025	2030	2035	2040	2045	2050
CL-N	[€/kWh]	0.136	0.162	0.188	0.217	0.242	0.266	0.284	0.298
CL-CN	[€/kWh]	0.154	0.181	0.209	0.232	0.258	0.277	0.293	0.308
CL-C	[€/kWh]	0.140	0.165	0.191	0.219	0.247	0.270	0.284	0.299
CL-S	[€/kWh]	0.159	0.185	0.213	0.237	0.261	0.280	0.294	0.309
CL-A	[€/kWh]	0.196	0.222	0.242	0.266	0.287	0.301	0.317	0.333
CL-M	[€/kWh]	0.142	0.169	0.196	0.222	0.243	0.267	0.288	0.303

[1] These values are the average annual prices of the year 2018, obtained from the website of each retail company that operates as an electricity distributor in the Chilean regional capitals.

Table S1.6 Electricity retail prices for industrial consumers by zone.

Zone	Unit	2015[1]	2020	2025	2030	2035	2040	2045	2050
CL-N	[€/kWh]	0.110	0.137	0.159	0.185	0.214	0.248	0.270	0.284
CL-CN	[€/kWh]	0.127	0.151	0.175	0.203	0.236	0.259	0.276	0.290
CL-C	[€/kWh]	0.122	0.145	0.169	0.195	0.227	0.260	0.273	0.287
CL-S	[€/kWh]	0.132	0.155	0.180	0.208	0.241	0.265	0.278	0.292
CL-A	[€/kWh]	0.151	0.175	0.203	0.236	0.263	0.276	0.290	0.305
CL-M	[€/kWh]	0.106	0.133	0.154	0.179	0.207	0.240	0.268	0.281

[1] These values are the average annual prices of the year 2018, obtained from the website of each retail company that operates as an electricity distributor in the Chilean regional capitals.

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Section S2. Results detail.

Table S2.1 Installed power capacity in GW – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.29	3.29	3.25	1.52	0.38	0.01	0.01
OCGT	0.63	0.61	0.60	0.60	0.35	0.04	0.01	0.4
Methane CHP	0.01	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.24	0.73	1.34	1.41	1.66	1.68	1.70
Biomass CHP	0.33	0.33	0.32	0.24	0.20	0	0	0
Waste-to-energy CHP	0.00	0.05	0.10	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.05	0.09	0.11	0.11	0.48	0.66	0.68
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	0.58	0.58	0.58	0	0
PV single-axis	0	0	3.25	3.25	3.25	3.25	14.56	21.82
PV prosumers	0	1.56	4.87	8.45	16.64	21.87	27.02	30.48
Wind onshore	0.91	7.35	8.27	8.27	8.09	7.36	2.68	1.77
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.11	3.11	3.11	3.11	3.11	3.11	3.10
Hydro reservoir (dam)	3.33	3.33	3.33	3.33	3.33	3.33	3.33	3.33
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.2 Installed power capacity in GW – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.56	3.56	3.52	1.79	0.65	0.27	0.27
OCGT	0.63	0.61	0.60	0.60	0.35	0.04	0.01	0
Methane CHP	0.01	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.10	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.20	0	0	0
Waste-to-energy CHP	0	0.05	0.1	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.05	0.06	0.21	0.34	0.45	0.52	0.96
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	8.01	20.17	38.42	78.65	139.34
PV single-axis	0	0	8.63	18.52	18.52	18.52	18.52	18.52
PV prosumers	0	1.56	5.04	8.94	17.15	22.56	27.87	31.42
Wind onshore	0.91	7.86	21.57	21.57	21.39	20.66	13.71	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.11	3.11	3.11	3.11	3.11	3.10	3.10
Hydro reservoir (dam)	3.33	3.33	3.33	3.63	3.63	3.63	3.63	3.63
Coal PP hard coal	4.32	4.20	3.89	3.89	3.89	3.30	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.3 Installed power capacity in GW – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.57	3.57	3.53	1.8	0.66	0.28	0.28
OCGT	0.63	0.61	0.6	0.6	0.35	0.04	0.01	0
Methane CHP	0.01	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0.00	0.05	0.1	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.05	0.06	0.25	0.35	0.47	0.52	0.91
Geothermal electricity	0	0	0	0	0	0	0	1.47
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	15.89	33.25	63.81	92.17	139.94
PV single-axis	0.00	0.00	6.66	16.00	16.00	20.95	36.65	36.66
PV prosumers	0.00	1.56	5.04	8.94	17.15	22.56	27.87	31.42
Wind onshore	0.91	7.88	23.58	23.58	23.4	22.67	15.7	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.11	3.11	3.11	3.11	3.11	3.1	3.11
Hydro reservoir (dam)	3.33	3.33	3.33	4.06	4.06	4.06	4.06	4.06
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.4 Installed power capacity in GW – 1-node power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.64	3.64	3.6	1.87	0.73	0.35	0.35
OCGT	0.63	0.61	0.6	0.6	0.35	0.04	0.01	0
Methane CHP	0.01	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.1	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.05	0.06	0.24	0.35	0.48	0.51	0.91
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	16.58	35.59	66.23	96.99	168.34
PV single-axis	0.00	0.00	6.19	16.6	16.6	22.76	36.55	36.55
PV prosumers	0.00	1.56	5.04	8.94	17.15	22.56	27.87	31.42
Wind onshore	0.91	7.97	24.28	24.28	24.1	23.37	16.31	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.11	3.11	3.11	3.11	3.11	3.11	3.1
Hydro reservoir (dam)	3.33	3.33	3.33	4.34	4.5	4.5	4.5	4.5
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.5 Installed power capacity in GW – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.31	3.31	3.27	1.68	0.66	0.49	0.64
OCGT	0.63	0.62	0.61	0.61	0.36	0.05	0.02	0.44
Methane CHP	0.01	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.39	0.74	1.33	1.41	1.69	1.71	1.77
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.1	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.05	0.1	0.2	0.29	0.37	0.39	0.51
Geothermal electricity	0.00	0.72	0.72	1.76	1.78	1.78	1.96	1.96
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	0.58	0.58	1.52	2.78	14.72
PV single-axis	0	7.36	11.83	13.82	14.33	15.16	15.97	11.75
PV prosumers	0	1.75	5.41	9.42	17.24	22.69	28.04	31.86
Wind onshore	0.91	0.91	0.91	0.91	0.73	0	0	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.11	3.11	3.1	3.11	3.1	3.1	3.1
Hydro reservoir (dam)	3.33	3.9	3.9	3.9	3.9	3.9	3.9	3.95
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.6 Installed power capacity in GW – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.3	3.3	3.26	1.53	0.39	0.01	0.01
OCGT	0.63	0.66	1.38	3.24	2.99	2.68	2.65	2.65
Methane CHP	0.01	0.43	0.43	0.43	0.43	0.43	0.43	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.25	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.15	0.17	0.18	0.18	0.16
Biogas CHP	0.05	0.16	0.17	0.18	0.21	0.32	0.46	0.72
Geothermal electricity	0	1.00	1.23	1.63	1.63	1.63	1.63	1.63
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	1.41	22.65	39.33	52.12	67.86	86.7	131.26
PV single-axis	0	15.84	20.69	27.82	29.43	31.92	33.8	18.71
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	0.97	1.2	1.61	2.09	2.19	2.4	2.64
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.11	3.11	3.11	3.11	3.1
Hydro reservoir (dam)	3.33	4.12	4.75	4.75	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.7 Installed power capacity in GW – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.3	3.3	3.26	1.54	0.4	0.02	0.03
OCGT	0.63	0.61	1.71	4.5	4.25	3.93	3.9	3.9
Methane CHP	0.01	0.38	0.38	0.37	0.37	0.37	0.37	0.01
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.16	0.18	0.18	0.18	0.15
Biogas CHP	0.05	0.17	0.17	0.18	0.22	0.37	0.47	0.69
Geothermal electricity	0	0.48	1.5	1.5	2.08	2.08	2.08	2.08
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	1.17	23.76	50.18	67.78	104	122.45	168.7
PV single-axis	0	15.89	20.38	29.11	32.16	34.07	40.66	29.05
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	0.97	1.24	1.72	2.92	2.8	3.01	3.25
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.11	3.11	3.11
Hydro reservoir (dam)	3.33	4.12	4.99	4.99	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.8 Installed power capacity in GW – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.31	3.31	3.27	1.54	0.4	0.02	0.02
OCGT	0.63	0.61	1.86	5.93	5.68	5.37	5.34	5.33
Methane CHP	0.01	0.48	0.48	0.48	0.47	0.47	0.47	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.25	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.16	0.19	0.19	0.19	0.16
Biogas CHP	0.05	0.16	0.17	0.18	0.22	0.4	0.57	0.81
Geothermal electricity	0	0	0.38	0.94	0.94	0.95	1.57	1.8
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	1.44	25.19	50.37	72.46	105.73	123.01	166.96
PV single-axis	0	17.4	22.73	33.81	36.83	41.05	44.46	34.49
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	0.99	1.25	1.72	3.62	3.43	3.64	3.88
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.1	3.11	3.11
Hydro reservoir (dam)	3.33	4.12	4.99	4.99	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.9 Installed power capacity in GW – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.29	3.29	3.25	1.53	0.46	0.16	0.18
OCGT	0.63	0.65	0.64	0.64	0.39	0.08	0.05	0.61
Methane CHP	0	0.01	0.01	0	0	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.39	0.75	1.35	1.42	1.66	1.68	1.7
Biomass CHP	0.32	0.32	0.31	0.24	0.19	0	0	0
Waste-to-energy CHP	0	0.05	0.1	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.06	0.13	0.22	0.42	0.5	0.54	0.64
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	0.58	0.58	0.58	0.58	0	0
PV single-axis	0	6.26	20.62	20.62	20.62	21.44	23.56	27.17
PV prosumers	0	1.75	5.41	9.42	17.24	22.69	28.04	31.86
Wind onshore	0.91	0.91	0.91	0.91	0.73	0	0	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Hydro reservoir (dam)	3.33	3.35	4.12	4.12	4.12	4.12	4.12	4.12
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.10 Installed power capacity in GW – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.3	3.3	3.26	1.53	0.39	0.01	0.01
OCGT	0.63	1.01	4.86	4.86	4.61	4.3	4.27	4.26
Methane CHP	0.01	0.01	0.01	0.01	0.01	0	0	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.13	0.14	0.17	0.18	0.2	0.28	0.56
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	4.37	23.44	38.35	50.87	68.21	98.46
PV single-axis	0	6.13	36.92	42.94	43.03	45.28	46.58	41.46
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	1.05	2.02	2.31	2.63	2.4	2.59	2.37
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Hydro reservoir (dam)	3.33	4.12	4.99	4.99	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.11 Installed power capacity in GW – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.3	3.3	3.27	1.54	0.4	0.02	0.02
OCGT	0.63	1.08	6.79	6.79	6.54	6.23	6.19	6.19
Methane CHP	0.01	0.1	0.1	0.1	0.09	0.09	0.09	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.13	0.13	0.16	0.18	0.23	0.32	0.55
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	8.79	28.48	49.56	70.55	90.39	126.82
PV single-axis	0	6.35	38.49	47.42	50.52	59.57	63.11	58.62
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	1.08	3.25	4.4	4.7	4.77	4.84	3.29
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Hydro reservoir (dam)	3.33	4.12	4.99	4.99	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.12 Installed power capacity in GW – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	3.29	3.67	3.67	3.63	1.9	0.76	0.38	0.38
OCGT	0.63	1.05	6.95	6.95	6.7	6.38	6.35	6.34
Methane CHP	0.01	0.27	0.27	0.26	0.26	0.26	0.26	0
ICE	2.83	2.69	2.5	2.49	2.24	0.19	0	0
Oil CHP	0.04	0.04	0.03	0.03	0.03	0	0	0
Biomass solid	0.11	0.11	0.1	0.08	0.07	0	0	0
Biomass CHP	0.33	0.33	0.32	0.24	0.2	0	0	0
Waste-to-energy CHP	0	0.05	0.11	0.15	0.15	0.15	0.15	0.15
Biogas CHP	0.05	0.15	0.16	0.17	0.18	0.2	0.21	0.4
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	0.58	0.58	8.47	25.37	51.03	72.86	93.45	127.8
PV single-axis	0	7.02	41.53	50.96	53.77	64.99	68.59	64.03
PV prosumers	0	0.88	3.92	8.03	17.28	23.22	29.06	32.81
Wind onshore	0.91	1.07	2.95	4.4	4.64	4.7	4.8	3.55
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	3.11	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Hydro reservoir (dam)	3.33	4.12	4.99	4.99	4.99	4.99	4.99	4.99
Coal PP hard coal	4.32	4.2	3.89	3.89	3.89	3.3	2.54	2.54
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.13 Electricity generation in TWh – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	10.38	2.88	2.88	2.85	1.33	0.7	0.05	0.02
OCGT	1.63	0.53	0.53	0.53	0.31	0.03	0	0.06
Methane CHP	0.01	0.01	0	0	0	0	0	0
ICE	2.16	1.35	0	0	0	0	0	0
Oil CHP	0.03	0.02	0	0	0	0	0	0
Biomass solid	0.11	1.98	6.08	11.15	11.73	13.84	13.99	14.14
Biomass CHP	0.33	2.72	2.64	2.03	1.67	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.05	0.35	0.62	0.94	0.03	1.39	3.8	3.79
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.1	1.1	1.1	1.1	1.1	1.1	0	0
PV single-axis	0	0	7.16	7.16	7.16	7.16	32.05	48.03
PV prosumers	0	2.99	9.31	16.16	31.83	41.85	51.69	58.32
Wind onshore	3.68	33.06	37.24	37.23	36.55	33.56	12.23	8.06
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.58	11.58	11.58	11.58	11.58	11.58	11.58	11.56
Hydro reservoir (dam)	12.41	12.41	12.41	12.41	12.41	12.41	12.41	12.41
Coal PP hard coal	28.62	8.66	0	0.14	0.11	0.4	0.06	0
Coal CHP	0.01	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.14 Electricity generation in TWh – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	10.38	3.12	3.12	3.08	1.11	0.06	0.02	0
OCGT	1.63	0.53	0.53	0.53	0.22	0	0	0
Methane CHP	0.02	0.02	0	0	0	0	0	0
ICE	2.16	1.35	0	0	0	0	0	0
Oil CHP	0.03	0.02	0	0	0	0	0	0
Biomass solid	0	0	0	0	0	0	0	0
Biomass CHP	0.25	0.03	0.01	0.02	0.06	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.13	0.4	0.32	0.61	0.69	0.83	0.87	0.88
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.1	1.1	1.1	15.33	38.6	73.51	150.48	266.58
PV single-axis	0	0	18.99	40.78	40.78	40.78	40.78	40.78
PV prosumers	0	2.99	9.64	17.1	32.8	43.17	53.32	60.12
Wind onshore	3.68	35.38	97.89	97.89	97.21	94.22	62.51	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.58	11.58	11.58	11.58	11.58	11.58	11.58	11.58
Hydro reservoir (dam)	12.41	12.41	12.41	13.54	13.54	13.54	13.54	13.54
Coal PP hard coal	28.62	22.09	0.11	0.27	0.7	0.88	0.34	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.15 Electricity generation in TWh – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	10.38	3.13	3.13	3.09	1.45	0.06	0.02	0
OCGT	1.63	0.53	0.53	0.53	0.28	0	0	0
Methane CHP	0.02	0.02	0	0	0	0	0	0
ICE	2.16	1.35	0	0	0	0	0	0
Oil CHP	0.03	0.02	0	0	0	0	0	0
Biomass solid	0	0	0	0	0	0	0	0
Biomass CHP	1	0.04	0.02	0.03	0.08	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.13	0.39	0.32	0.61	0.69	0.83	0.87	0.87
Geothermal electricity	0	0	0	0	0	0	0	12.21
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.1	1.1	1.1	30.4	63.62	122.08	176.34	267.74
PV single-axis	0	0	14.66	35.22	35.22	46.12	80.68	80.69
PV prosumers	0	2.99	9.64	17.1	32.8	43.17	53.32	60.12
Wind onshore	3.68	35.44	107.03	107.02	106.35	103.36	71.59	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.58	11.58	11.58	11.58	11.58	11.58	11.58	11.58
Hydro reservoir (dam)	12.41	12.41	12.41	15.15	15.15	15.15	15.15	15.15
Coal PP hard coal	28.62	24.92	0.26	0.45	0.83	0.9	0.45	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.16 Electricity generation in TWh – 1-node power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	10.38	3.18	3.18	3.15	1.52	0.05	0.03	0
OCGT	1.63	0.53	0.53	0.53	0.29	0	0	0
Methane CHP	0.02	0.02	0	0	0	0	0	0
ICE	2.16	1.35	0	0	0	0	0	0
Oil CHP	0.03	0.02	0	0	0	0	0	0
Biomass solid	0.25	0	0	0	0	0	0	0
Biomass CHP	1.77	0.04	0.02	0.03	0.08	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.19	0.39	0.32	0.61	0.69	0.83	0.87	0.86
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.1	1.1	1.1	31.73	68.08	126.72	185.57	322.08
PV single-axis	0	0	13.63	36.54	36.54	50.11	80.46	80.46
PV prosumers	0	2.99	9.64	17.1	32.8	43.17	53.32	60.12
Wind onshore	3.68	35.86	110.24	110.23	109.55	106.56	74.37	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.58	11.58	11.58	11.58	11.58	11.58	11.58	11.58
Hydro reservoir (dam)	12.41	12.41	12.41	16.18	16.8	16.8	16.8	16.8
Coal PP hard coal	28.62	25.8	0.31	0.47	0.86	0.96	0.43	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.17 Electricity generation in TWh – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	5.74	3.52	3.84	4.09	1.78	1.01	0.97	0.75
OCGT	2.68	0.54	0.53	0.53	0.31	0	0	0.01
Methane CHP	0.04	0.01	0	0	0	0	0	0
ICE	3.21	1.35	0	0	0	0	0	0
Oil CHP	0.06	0.02	0	0	0	0	0	0
Biomass solid	0.33	2.7	6.12	11.11	11.71	13.85	14	14.19
Biomass CHP	0.09	2.15	2.64	2.03	1.67	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.26	0.29	0.69	1.25	1.82	2.21	2.38	2.6
Geothermal electricity	0	5.96	5.96	14.65	14.83	14.84	16.34	16.34
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	1.2	1.2	1.2	1.2	3.17	5.4	26.86
PV single-axis	0	15.13	25.83	30.66	31.88	33.57	35.46	27.52
PV prosumers	0	3.24	9.93	17.26	31.5	41.44	51.2	58.15
Wind onshore	2.2	2.2	2.2	2.19	1.76	0	0	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.47	14.63	14.63	14.63	14.63	14.63	14.63	14.83
Coal PP hard coal	32.7	19.82	10.91	0	0	0.3	0.29	0
Coal CHP	0	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.18 Electricity generation in TWh – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	13.48	3.14	1.65	1.64	0.72	0.01	0.01	0.01
OCGT	3.02	0.33	0.69	1.62	1.39	0.01	0.01	0
Methane CHP	0.06	3.05	0.34	0.18	0.36	0.39	0.22	0
ICE	0.48	0.02	0	0	0	0	0	0
Oil CHP	0	0	0	0	0	0	0	0
Biomass solid	0.71	0	0	0	0	0	0	0
Biomass CHP	1.82	0.07	0.02	0.02	0.01	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.24	0.85	0.49	0.77	0.77	0.8	0.98	1.31
Geothermal electricity	0	8.33	10.25	13.54	11.89	11.7	11.88	11.89
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	2.64	38.13	68.34	90.87	118.4	151.27	228.41
PV single-axis	0	30.48	42.03	59.52	63.48	69.58	74.21	45.56
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	2.49	3.86	6.2	9.52	11.15	12.32	13.7
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.4	15.48	17.8	17.8	18.7	18.7	18.7	18.7
Coal PP hard coal	23.64	19.47	9.26	0	0	0.09	0.04	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.19 Electricity generation in TWh – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	9.12	3.5	1.66	1.64	0.74	0.02	0.01	0.02
OCGT	2.21	0.3	0.85	2.25	1.99	0.02	0.01	0.01
Methane CHP	0.04	2.66	0.26	0.48	0.44	0.3	0.16	0.01
ICE	0.74	0	0	0	0	0	0	0
Oil CHP	0.06	0	0	0	0	0	0	0
Biomass solid	0.71	0	0	0	0	0	0	0
Biomass CHP	1.83	0.06	0.01	0.01	0.01	0	0	0.01
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.24	0.87	0.49	0.77	0.78	0.84	0.98	1.23
Geothermal electricity	0	4.02	12.45	12.45	15.79	15.81	15.81	15.82
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	2.24	39.96	87.37	117.65	183.42	214.31	294.19
PV single-axis	0	31.05	41.67	63.11	70.58	75.27	91.04	70.38
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	2.51	4.06	6.78	12.02	13.48	14.65	16.02
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.44	15.48	18.7	18.7	18.7	18.7	18.7	18.7
Coal PP hard coal	28.57	26.11	11.56	0	0	0.2	0.06	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.20 Electricity generation in TWh – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	15.14	4.3	1.66	1.65	0.73	0.02	0.02	0.01
OCGT	3.4	0.3	0.93	2.96	2.63	0.02	0.01	0
Methane CHP	0.07	3.39	0.53	0.84	0.6	0.4	0.29	0
ICE	3.22	0	0	0	0	0	0	0
Oil CHP	0.2	0	0	0	0	0	0	0
Biomass solid	0.82	0	0	0	0	0	0	0
Biomass CHP	2.15	0.06	0.01	0.02	0.02	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.25	0.87	0.49	0.78	0.78	0.86	1.12	1.31
Geothermal electricity	0	0	3.17	7.8	7.79	7.88	13.06	14.94
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	2.69	42.42	86.91	126.64	186.03	216.02	291.11
PV single-axis	0	33.99	46.66	73.85	81.26	91.62	99.7	83.61
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	2.58	4.1	6.78	13.59	14.9	16.06	17.44
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.47	15.48	18.7	18.7	18.7	18.7	18.7	18.7
Coal PP hard coal	19.73	26.43	17.13	0	0	0.33	0.07	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.21 Electricity generation in TWh – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	9.09	2.35	3.12	3.39	0.44	0.39	0.28	0.24
OCGT	2.18	0.32	0.32	0.56	0.34	0.01	0	0.02
Methane CHP	0	0	0	0	0	0	0	0
ICE	2.16	0	0	0	0	0	0	0
Oil CHP	0.06	0	0	0	0	0	0	0
Biomass solid	0.89	3.24	6.21	11.2	11.81	13.84	13.99	14.17
Biomass CHP	1.1	1.65	2.58	1.96	1.6	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.26	0.41	0.83	1.69	2.7	3.07	3.28	3.46
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	1.2	1.2	1.2	1.2	1.2	0	0
PV single-axis	0	13.09	43	43	43	45	50.21	61.06
PV prosumers	0	3.24	9.93	17.26	31.5	41.44	51.2	58.15
Wind onshore	2.2	2.2	2.21	2.2	1.77	0.01	0.01	0
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.43
Hydro reservoir (dam)	12.47	12.57	15.48	15.48	15.48	15.48	15.48	15.48
Coal PP hard coal	28.62	28.46	0	0	0	0.08	0.12	0
Coal CHP	0.01	0	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.22 Electricity generation in TWh – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	9.08	4.97	1.65	1.63	0.76	0.02	0.02	0.01
OCGT	2.18	0.5	2.43	2.43	2.25	0.01	0.01	0
Methane CHP	0.04	0.09	0.01	0	0	0	0	0
ICE	2.16	0	0	0	0	0	0	0
Oil CHP	0.06	0	0	0	0	0	0	0
Biomass solid	0.6	0	0	0	0	0	0	0
Biomass CHP	1.81	0.44	0.02	0.01	0.01	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.18	0.88	0.49	0.73	0.75	0.73	0.85	1.01
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	1.2	7.76	42.67	73.8	99.45	132.99	190.61
PV single-axis	0	12.72	75.62	90.41	90.64	96.14	99.33	89.11
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	2.96	6.98	8.65	11.09	12.17	13.21	13.35
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.46	15.48	18.7	18.7	18.7	18.7	18.7	18.7
Coal PP hard coal	26.69	34.98	4.91	0	0	0.19	0.14	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.23 Electricity generation in TWh – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	9.08	8.44	1.65	1.63	0.76	0.01	0.01	0.01
OCGT	2.18	0.54	3.39	3.39	3.21	0.02	0.01	0
Methane CHP	0.04	0.76	0.13	0	0	0.05	0.03	0
ICE	2.16	0	0	0	0	0	0	0
Oil CHP	0.06	0	0	0	0	0	0	0
Biomass solid	0.6	0	0	0	0	0	0	0
Biomass CHP	1.81	0.84	0.02	0.01	0.01	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.17	0.88	0.49	0.74	0.76	0.78	0.88	0.99
Geothermal electricity	0	0	0	0	0	0	0	0
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	1.2	15.4	51.26	95.12	138.64	177.46	247.38
PV single-axis	0	13.26	79.85	101.77	109.37	131.57	140.26	131.55
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	3.1	10.36	16.37	18.72	21.53	22.04	18.31
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.46	15.48	18.7	18.7	18.7	18.7	18.7	18.7
Coal PP hard coal	28.6	34.98	2.4	0	0	0.29	0.22	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.24 Electricity generation in TWh – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
ST others	0	0	0	0	0	0	0	0
CCGT	9.08	8.1	1.83	1.81	0.89	0.02	0.02	0.01
OCGT	2.18	0.53	3.47	3.47	3.09	0.01	0.01	0
Methane CHP	0.04	1.93	0.23	0.12	0.06	0.14	0.08	0
ICE	2.16	0	0	0	0	0	0	0
Oil CHP	0.06	0	0	0	0	0	0	0
Biomass solid	0.6	0	0	0	0	0	0	0
Biomass CHP	1.81	0.94	0.02	0.01	0.01	0	0	0
Waste-to-energy CHP	0	0.42	0.85	1.28	1.28	1.28	1.28	1.28
Biogas CHP	0.17	0.88	0.49	0.76	0.75	0.75	0.77	0.95
Geothermal electricity	0	0	0.01	0.01	0.01	0.02	0.02	0.02
CSP ST	0	0	0	0	0	0	0	0
PV fixed tilted	1.2	1.2	14.86	46.49	99.93	144.22	184.12	250.62
PV single-axis	0	14.68	86.12	109.26	116.16	143.69	152.53	143.89
PV prosumers	0	1.64	7.14	14.47	31.4	42.17	52.76	59.56
Wind onshore	2.2	3.08	9.58	16.92	18.9	21.68	22.32	19.44
Wind offshore	0	0	0	0	0	0	0	0
Hydro run-of-river	11.45	11.45	11.45	11.45	11.45	11.45	11.45	11.45
Hydro reservoir (dam)	12.46	15.48	18.7	18.7	18.7	18.7	18.7	18.7
Coal PP hard coal	28.6	34.92	2.11	0	0	0.27	0.14	0
Coal CHP	0.01	0.01	0	0	0	0	0	0
Nuclear PP	0	0	0	0	0	0	0	0

Table S2.25 Installed power transmission capacity in GW-km – 6-nodes-interconnected power, heat, transport and desalination sectors.

Zone	2015	2020	2025	2030	2035	2040	2045	2050
New	0	53	1,174	548	1,419	777	372	589
Total	5,747	5,800	6,974	7,521	8,940	9,717	10,089	10,678

Table S2.26 Curtailment in percent by scenario.

Scenario	2015	2020	2025	2030	2035	2040	2045	2050
1-node P	0.2	0.1	2.9	5.2	5	4.5	3.9	4.3
1-node PH	0	0	1.6	3.2	2.4	2.7	3.4	4.2
1-node PHT	0	0	1.6	2.6	2.5	3.6	3.9	4.9
1-node PHTD	1	0	1.6	2.6	2.5	3.6	3.9	5.0
6-nodes-isolated P	0.6	1.4	4.9	8.1	10.1	9.6	9.7	11.6
6-nodes-isolated PH	0.6	0.2	2.4	3.9	6.1	7.9	7.4	7.9
6-nodes-isolated PHT	1.1	0.1	2.2	4.1	6.3	8	8	8.7
6-nodes-isolated PHTD	1	0	2.6	4.4	7.6	8.8	8.3	8.8
6-nodes-interconnected P	0.6	0.1	2.1	3.2	4.4	4.1	3.9	3.9
6-nodes-interconnected PH	2.1	0	0.5	4	5	4.4	4.4	4.8
6-nodes-interconnected PHT	2.7	0	0.7	4.3	8.9	8.8	8.1	7.8
6-nodes-interconnected PHTD	2	0	0.7	4.1	8.9	9.1	8.4	7.7

Table S2.27 Storage capacity in GWh – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	3.33	7.95	13.73	25.07	33.11	40.83	46.23
Battery utility	0	0	0	0	3.12	10.91	46.91	74.98
PHES	0	0	5.81	5.81	18.95	20.35	23.72	25.46
TES (MT heat)	0	0	0	0	0	0	0	0
DH (LT heat)	0	0	0	0	0	0	0	0
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	10.76	141.93	310.72	310.72	310.72	310.72	310.72

Table S2.28 Storage capacity in GWh – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	3.33	8.38	14.84	27.27	36.12	44.59	50.38
Battery utility	0	0	0	0.34	1.40	2.07	36.06	144.23
PHES	0	0	0	0	15.15	49.43	62.77	62.77
TES (MT heat)	0	0	24.87	88.88	112.45	132.69	158.57	185.60
DH (LT heat)	0	6.40	20.77	21.21	30.01	39.61	37.08	42.91
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.32	0.53	0.75	80.79	549.80	878.36	878.36

Table S2.29 Storage capacity in GWh – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	3.33	8.38	14.84	27.27	36.12	44.59	50.38
Battery utility	0	0	0	0.16	0.16	0.34	38.77	159.84
PHES	0	0	0	9.99	49.12	88.86	90.02	90.02
TES (MT heat)	0	0	20.77	90.15	114.28	135.14	160.37	190.83
DH (LT heat)	0	6.64	21.20	24.09	33.36	55.71	55.15	60.52
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.24	0.52	0.77	14.64	309.98	1769.68	1769.68

Table S2.30 Storage capacity in GWh – 1-node power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	3.33	8.38	14.84	27.27	36.12	44.59	50.38
Battery utility	0	0	0	0.05	0.05	0.14	40.83	166.26
PHES	0	0	0	13.27	56.05	99.33	100.18	100.18
TES (MT heat)	0	0	19.12	90.68	119.19	139.39	165.48	192.25
DH (LT heat)	0	6.65	21.16	24.18	32.46	54.77	54.11	62.81
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.18	0.56	0.76	27.14	351.81	1715.67	1715.67

Table S2.31 Storage capacity in GWh – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	2.84	7.69	13.64	25.33	33.73	41.72	47.46
Battery utility	0	0	0.11	0.11	0.98	11.55	23.19	37.16
PHES	0	14.24	32.69	48.22	55.44	58.48	58.74	92.44
TES (MT heat)	0	0.00	0.00	0.00	0.00	0.00	0.01	0.01
DH (LT heat)	0	0.02	0.16	0.16	0.41	0.41	0.40	0.63
A-CAES	0	0.58	0.58	0.58	0.58	0.58	0.58	0.58
Gas storage	0	49.31	164.49	212.64	880.16	1149.50	1159.68	1483.50

Table S2.32 Storage capacity in GWh – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0.04	0.06	0.10	0.11	12.27	30.01	61.82
PHES	0	27.60	61.54	109.98	123.95	135.24	135.32	136.16
TES (MT heat)	0	0.31	49.53	97.03	127.77	142.96	160.91	182.96
DH (LT heat)	0	9.48	30.54	33.31	39.76	48.46	48.06	47.74
A-CAES	0	0.04	0.04	0.08	0.08	0.08	0.08	0.08
Gas storage	0	51.47	62.10	100.59	175.00	1101.39	2164.37	2419.29

Table S2.33 Storage capacity in GWh – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0	0.01	0.01	0.03	10.21	22.81	56.98
PHES	0	23.38	66.15	141.92	167.30	180.32	181.49	183.69
TES (MT heat)	0	0.07	50.13	100.78	125.53	143.75	158.97	181.90
DH (LT heat)	0	12.18	31.55	34.03	42.63	64.14	68.71	74.98
A-CAES	0	0	0	0	0	0.14	0.15	0.29
Gas storage	0	1.00	13.33	47.98	317.41	1417.73	2976.63	3761.20

Table S2.34 Storage capacity in GWh – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0	0.04	0.09	0.12	17.74	26.66	57.87
PHES	0	31.24	76.48	159.40	190.33	203.68	204.72	209.57
TES (MT heat)	0	0.07	50.04	100.41	127.58	141.79	156.13	179.44
DH (LT heat)	0	12.08	31.21	33.88	44.07	64.11	63.50	70.61
A-CAES	0	0	0	0.04	0.04	0.04	0.08	0.08
Gas storage	0	0.24	11.77	91.85	543.92	1694.65	2609.99	3535.45

Table S2.35 Storage capacity in GWh – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	2.84	7.69	13.64	25.33	33.73	41.72	47.46
Battery utility	0	0	0	0	0.01	9.88	20.86	39.57
PHES	0	0.61	74.52	74.52	77.90	80.18	81.17	81.80
TES (MT heat)	0	0	0	0	0	0	0	0
DH (LT heat)	0	0	0	0	0.84	0.84	0.84	0.84
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	4.34	242.97	295.83	479.09	559.39	641.49	663.28

Table S2.36 Storage capacity in GWh – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0	0	0	0	4.97	26.97	59.73
PHES	0	0.87	76.37	120.91	136.94	152.86	152.86	154.90
TES (MT heat)	0	0.13	49.02	101.44	124.69	146.15	160.87	185.19
DH (LT heat)	0	12.78	31.48	33.50	38.83	48.56	43.80	41.86
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.04	6.91	27.69	85.52	740.77	1590.59	2043.49

Table S2.37 Storage capacity in GWh – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0	0	0	0	16.89	35.73	79.86
PHES	0	1.40	98.95	152.35	180.82	188.69	188.69	190.67
TES (MT heat)	0	0.05	55.52	99.44	120.15	131.18	148.45	174.95
DH (LT heat)	0	15.11	34.93	37.51	45.92	62.85	58.39	59.65
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.20	9.09	20.17	145.61	992.27	2140.98	2831.96

Table S2.38 Storage capacity in GWh – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.58	7.82	14.20	26.93	36.51	45.80	52.17
Battery utility	0	0	0	0	0	17.40	39.59	81.16
PHES	0	1.19	110.00	151.33	182.76	195.89	195.89	197.35
TES (MT heat)	0	0.05	56.18	94.97	118.87	129.06	147.79	173.15
DH (LT heat)	0	14.56	31.23	32.85	41.37	57.88	59.75	62.90
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.40	8.12	31.81	509.61	1513.84	2869.53	3571.24

Table S2.39 Storage output in TWh – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	1.14	2.65	4.49	8.19	10.85	13.40	15.12
Battery utility	0	0	0	0	15.62	6.51	20.68	37.58
PHES	0	0	1.52	1.63	9.92	5.47	5.97	5.93
TES (MT heat)	0	0	0	0	0	0	0	0
DH (LT heat)	0	0	0	0	0	0	0	0
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	2.38	4.23	6.13	4.69	2.01	0.09	0.17

Table S2.40 Storage output in TWh – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	1.14	2.78	4.83	8.88	11.81	14.58	16.44
Battery utility	0	0	0	0.38	0.98	1.18	14.41	57.73
PHES	0	0	0	0	2.95	10.10	14.73	15.59
TES (MT heat)	0	0	4.97	26.54	36.38	45.46	55.07	63.58
DH (LT heat)	0	2.01	6.01	7.14	9.60	12.82	12.21	14.46
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	1.31	4.23	6.17	14.32	21.22	32.33	44.23

Table S2.41 Storage output in TWh – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	1.14	2.78	4.83	8.88	11.81	14.58	16.44
Battery utility	0	0	0	0.17	0.15	0.19	14.73	64.26
PHES	0	0	0	2.18	11.24	21.77	21.69	23.15
TES (MT heat)	0	0	4.04	29.08	38.78	47.51	55.86	65.13
DH (LT heat)	0	2.11	6.24	8.05	11.11	18.55	18.69	20.96
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	1.31	4.23	6.33	8.22	14.29	27.54	39.55

Table S2.42 Storage output in TWh – 1-node power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	1.14	2.78	4.83	8.88	11.81	14.58	16.44
Battery utility	0	0	0	0.05	0.05	0.07	16.25	68.62
PHES	0	0	0	2.92	13.00	24.79	24.69	26.25
TES (MT heat)	0	0	3.67	29.40	40.36	48.76	57.29	65.73
DH (LT heat)	0	2.12	6.24	8.09	10.93	18.46	18.37	21.70
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	1.32	4.23	6.28	8.04	14.62	27.72	39.91

Table S2.43 Storage output in TWh – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.93	2.50	4.30	7.94	10.54	13.16	26.40
Battery utility	0	0	0.05	0.08	0.45	5.13	10.22	16.15
PHES	0	2.89	8.02	12.76	15.32	15.86	15.96	16.45
TES (MT heat)	0	0	0	0	0	0	0.00	0.00
DH (LT heat)	0	0.01	0.19	0.37	0.86	1.01	1.04	1.19
A-CAES	0	2.57	0.87	1.37	1.44	1.31	1.25	0.66
Gas storage	0	2.25	3.30	3.85	3.49	2.79	2.77	2.98

Table S2.44 Storage output in TWh – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0.07	0.04	0.09	0.07	5.24	12.51	25.35
PHES	0	6.14	15.72	28.24	32.94	37.51	37.83	37.39
TES (MT heat)	0	0.14	10.85	26.41	38.03	43.25	49.11	55.76
DH (LT heat)	0	4.25	11.44	12.60	13.43	16.96	15.82	15.59
A-CAES	0	0.01	0.00	0.02	0.02	0.02	0.02	0.02
Gas storage	0	0.26	3.91	5.83	8.38	14.15	24.74	35.14

Table S2.45 Storage output in TWh – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0.00	0.01	0.00	0.02	5.09	9.94	24.46
PHES	0	5.36	17.00	37.62	45.52	52.51	52.36	51.27
TES (MT heat)	0	0.02	11.00	27.56	37.61	46.05	49.00	56.49
DH (LT heat)	0	5.10	13.85	16.22	20.70	41.70	33.68	27.43
A-CAES	0	0	0	0	0	0.02	0.02	0.04
Gas storage	0	0.18	3.90	5.79	8.10	14.51	25.00	36.79

Table S2.46 Storage output in TWh – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0	0.02	0.09	0.09	7.67	12.11	25.13
PHES	0	6.94	20.12	42.49	53.13	59.98	60.19	59.02
TES (MT heat)	0	0.01	11.04	27.11	38.06	42.77	48.32	55.13
DH (LT heat)	0	5.28	13.93	19.79	23.65	45.58	32.78	27.96
A-CAES	0	0	0	0.01	0.01	0.01	0.02	0.02
Gas storage	0	0.17	3.90	5.82	8.49	13.79	22.00	35.06

Table S2.47 Storage output in TWh – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.93	2.50	4.30	7.94	10.54	13.16	26.40
Battery utility	0	0	0	0	0.01	4.92	10.16	19.66
PHES	0	0.28	17.03	17.96	19.74	20.03	20.29	20.31
TES (MT heat)	0	0	0	0	0	0	0	0
DH (LT heat)	0	0	0	0	1.20	0.95	1.02	1.32
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	2.19	3.63	4.93	2.90	1.93	1.72	1.65

Table S2.48 Storage output in TWh – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0	0	0	0	2.76	13.65	29.55
PHES	0	0.32	19.46	30.47	38.05	45.54	44.94	44.50
TES (MT heat)	0	0.04	8.47	25.87	51.01	67.47	64.36	62.45
DH (LT heat)	0	4.92	12.10	13.23	21.04	29.32	21.53	15.37
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.16	3.91	5.90	7.37	12.37	22.36	32.57

Table S2.49 Storage output in TWh – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0	0	0	0	7.68	16.01	34.79
PHES	0	0.47	25.70	39.97	51.62	57.26	56.33	55.91
TES (MT heat)	0	0.01	9.85	26.71	38.13	43.13	49.60	57.69
DH (LT heat)	0	6.08	13.73	14.59	19.18	41.04	26.18	23.26
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.16	3.91	5.87	6.67	12.52	22.03	33.54

Table S2.50 Storage output in TWh – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Battery prosumers	0	0.19	2.51	4.73	9.77	14.04	24.16	27.84
Battery utility	0	0	0	0	0.00	8.06	17.77	35.29
PHES	0	0.40	28.25	41.47	53.99	61.31	60.62	60.67
TES (MT heat)	0	0.01	10.04	24.66	36.82	41.42	48.88	56.96
DH (LT heat)	0	5.79	12.75	13.24	18.57	40.60	26.74	24.91
A-CAES	0	0	0	0	0	0	0	0
Gas storage	0	0.16	3.90	5.82	8.62	15.69	25.24	36.01

Table S2.51 Installed capacity for fuel conversion in GW_{output} – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0	0	0	0	0
Steam methane reforming	0	0	0	0	0	0	0	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0	0	0	0	0
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.52 Installed capacity for fuel conversion in GW_{output} – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0.01	2.41	4.16	9.34	14.61
Steam methane reforming	0	0	0	0	0	0	0	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0.01	1.09	2.14	3.76	5.6
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.53 Installed capacity for fuel conversion in GW_{output} – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.14	1.32	3.28	10.63	19.45	20.98
Steam methane reforming	0	0.01	0.01	0.01	0.01	0.01	0.01	0
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0.05	0.31	1.26	3.52	3.52
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.35	0.71	2.21	2.59	2.59

Table S2.54 Installed capacity for fuel conversion in GW_{output} – 1-node power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.14	1.3	3.25	10.82	19.5	25.94
Steam methane reforming	0	0.01	0.01	0.01	0.01	0.01	0.01	0
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0.03	0.28	1.31	3.52	5.55
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.35	0.71	2.21	2.59	2.59

Table S2.55 Installed capacity for fuel conversion in GW_{output} – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0.05	0.18	0.41	0.75	0.75	0.75	0.7
Steam methane reforming	0	0.1	0.11	0.11	0.11	0.11	0.11	0.01
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0	0.17	0.2	0.2	0.27
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0.09	0.09	0.09	0.09	0.09	0.09	0

Table S2.56 Installed capacity for fuel conversion in GW_{output} – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0.3	1.03	2.69	7.28	11.23
Steam methane reforming	0	0.12	0.12	0.12	0.12	0.12	0.12	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0.01	0.46	1.47	3.28	5.05
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0.08	0.08	0.08	0.08	0.08	0.08	0

Table S2.57 Installed capacity for fuel conversion in GW_{output} – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.05	1.7	4.59	13.94	20.6	26.14
Steam methane reforming	0	0.01	0.02	0.02	0.02	0.03	0.03	0.03
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0	0.43	1.64	3.54	5.8
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.37	0.76	2.28	2.69	2.69

Table S2.58 Installed capacity for fuel conversion in GW_{output} – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.05	1.72	4.92	13.67	19.37	25.25
Steam methane reforming	0	0.01	0.02	0.02	0.02	0.02	0.03	0.02
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0.01	0.52	1.47	2.99	5.15
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.37	0.76	2.28	2.69	2.69

Table S2.59 Installed capacity for fuel conversion in GW_{output} – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0.18	0.88	0.88	0.88	0.88
Steam methane reforming	0	0	0	0	0	0	0	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0.14	0.24	0.24	0.24	0.24
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.60 Installed capacity for fuel conversion in GW_{output} – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0	0.45	2.26	6.24	9.96
Steam methane reforming	0	0	0	0	0	0	0	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0	0.25	1.16	2.93	4.61
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.61 Installed capacity for fuel conversion in GW_{output} – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.04	1.33	3.49	12.07	17.89	22.71
Steam methane reforming	0	0.01	0.03	0.03	0.03	0.03	0.03	0.02
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0	0.18	1.26	2.97	4.89
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.35	0.71	2.19	2.58	2.58

Table S2.62 Installed capacity for fuel conversion in GW_{output} – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.05	1.34	4.19	13.04	18.62	23.54
Steam methane reforming	0	0.01	0.03	0.03	0.03	0.03	0.03	0.02
Liquid hydrogen	0	0	0	0.01	0.06	0.22	0.51	0.92
Methanation	0	0	0	0	0.61	1.86	3.6	5.48
LNG	0	0.01	0.01	0.03	0.04	0.07	0.13	0.27
Fischer–Tropsch	0	0	0	0.35	0.71	2.19	2.58	2.58

Table S2.63 e-Fuel output in TWh – 1-node power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0	0	0	0	0
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0	0	0	0	0
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.64 e-Fuel output in TWh – 1-node power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0.05	9.97	18.75	32.27	46.75
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0.04	8.18	15.39	26.5	38.38
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.65 e-Fuel output in TWh – 1-node power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.37	5.47	14.16	43.4	66.66	67.36
Liquid hydrogen	0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation	0	0	0	0.2	2.09	8.45	21.71	23.95
LNG	0	0	0.01	0.1	0.3	0.54	1.12	2.28
Fischer–Tropsch	0	0	0	2.94	5.93	18.37	21.53	17.69

Table S2.66 e-Fuel output in TWh – 1-node power, heat, transport and desalination sectors.

Technology		2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis		0	0	0.37	5.41	13.93	43.82	66.89	85.19
Liquid hydrogen		0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation		0	0	0	0.15	1.91	8.79	21.9	34.04
LNG		0	0	0.01	0.1	0.3	0.54	1.12	2.28
Fischer–Tropsch		0	0	0	2.94	5.93	18.37	21.53	21.53

Table S2.67 e-Fuel output in TWh – 6-nodes-isolated power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0.36	0.46	1.19	2.31	2.57	2.61	2.17
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0.01	0.01	0.01	0.88	1.08	1.11	1.7
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0.74	0.74	0.74	0.75	0.75	0.75	0

Table S2.68 e-Fuel output in TWh – 6-nodes-isolated power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0.01	0	1.03	4.14	11.22	24.43	36.75
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0.01	0	0.03	2.59	8.4	19.25	30.17
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0.68	0.68	0.68	0.68	0.69	0.69	0

Table S2.69 e-Fuel output in TWh – 6-nodes-isolated power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.18	5.36	14.96	44.75	65.11	83.49
Liquid hydrogen	0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation	0	0	0	0	2.32	8.88	19.47	31.68
LNG	0	0	0.01	0.04	0.24	0.52	1.12	2.28
Fischer–Tropsch	0	0	0	3.1	6.3	18.98	22.39	22.39

Table S2.70 e-Fuel output in TWh – 6-nodes-isolated power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.18	5.41	15.46	43.94	61.85	81.54
Liquid hydrogen	0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation	0	0	0	0.05	2.73	8.21	16.8	30.09
LNG	0	0	0	0.05	0.23	0.55	1.12	2.28
Fischer–Tropsch	0	0	0	3.1	6.3	18.97	22.39	22.39

Table S2.71 e-Fuel output in TWh – 6-nodes-interconnected power sector.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	1.49	2.19	1.61	1.6	1.54
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	1.16	1.71	1.25	1.24	1.2
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.72 e-Fuel output in TWh – 6-nodes-interconnected power and heat sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0	0.01	1.8	7.93	20.24	32.82
Liquid hydrogen	0	0	0	0	0	0	0	0
Methanation	0	0	0	0.01	1.48	6.51	16.62	26.94
LNG	0	0	0	0	0	0	0	0
Fischer–Tropsch	0	0	0	0	0	0	0	0

Table S2.73 e-Fuel output in TWh – 6-nodes-interconnected power, heat and transport sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.12	5.16	12.49	41.04	59.9	77.51
Liquid hydrogen	0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation	0	0	0	0	0.8	6.68	16.31	27.88
LNG	0	0	0.01	0.04	0.19	0.55	1.12	2.28
Fischer–Tropsch	0	0	0	2.9	5.88	18.26	21.46	21.46

Table S2.74 e-Fuel output in TWh – 6-nodes-interconnected power, heat, transport and desalination sectors.

Technology	2015	2020	2025	2030	2035	2040	2045	2050
Water electrolysis	0	0	0.18	5.16	14.9	44.86	63.62	80.42
Liquid hydrogen	0	0	0	0.09	0.51	1.8	4.26	7.63
Methanation	0	0	0	0	2.78	9.81	19.36	30.27
LNG	0	0	0.01	0.04	0.23	0.55	1.12	2.28
Fischer–Tropsch	0	0	0	2.9	5.88	18.26	21.46	21.46

Table S2.75 CO_2 DAC capacity in $MtCO_2/a$ by scenario.

Scenario	2015	2020	2025	2030	2035	2040	2045	2050
1-node P	0	0	0	0	0	0	0	0
1-node PH	0	0	0	0.01	1.88	3.71	6.52	9.72
1-node PHT	0	0	0	0.99	2.34	7.79	12.67	13.01
1-node PHTD	0	0	0	0.98	2.3	7.88	12.67	16.2
6-nodes-isolated P	0	0.22	0.22	0.22	0.47	0.52	0.52	0.53
6-nodes-isolated PH	0	0.22	0.22	0.23	1	2.75	5.9	8.77
6-nodes-isolated PHT	0	0	0	0.99	2.72	8.63	12.98	16.89
6-nodes-isolated PHTD	0	0	0	1.01	2.87	8.34	12.01	15.76
6-nodes-interconnected P	0	0	0	0.22	0.37	0.37	0.37	0.37
6-nodes-interconnected PH	0	0	0	0	0.43	2.02	5.08	7.99
6-nodes-interconnected PHT	0	0	0	0.93	2.12	7.76	11.7	15.03
6-nodes-interconnected PHTD	0	0	0	0.93	2.86	8.8	12.81	16.07

Table S2.76 CO_2 DAC_{output} in $MtCO_2$ by scenario.

Scenario	2015	2020	2025	2030	2035	2040	2045	2050
1-node P	0	0	0	0	0	0	0	0
1-node PH	0	0	0	0.01	1.62	3.05	5.24	7.6
1-node PHT	0	0	0	0.94	2.22	7.28	10.87	10.14
1-node PHTD	0	0	0	0.93	2.19	7.34	10.9	13.31
6-nodes-isolated P	0	0.21	0.21	0.21	0.37	0.4	0.41	0.3
6-nodes-isolated PH	0	0.21	0.21	0.22	0.72	1.87	4.02	5.97
6-nodes-isolated PHT	0	0	0	0.95	2.38	7.55	10.68	13.1
6-nodes-isolated PHTD	0	0	0	0.95	2.46	7.41	10.15	12.78
6-nodes-interconnected P	0	0	0	0.21	0.3	0.22	0.22	0.21
6-nodes-interconnected PH	0	0	0	0	0.29	1.29	3.29	5.33
6-nodes-interconnected PHT	0	0	0	0.88	1.95	6.89	9.77	12.06
6-nodes-interconnected PHTD	0	0	0	0.88	2.34	7.51	10.38	12.54